

Computer Vision Project Pitch

Objective

The purpose of this project is to help boaters detect boating hazards from their iPhone camera. The goal for this project is to analyze maritime images, and identify potential boating hazards. Some example hazards are buoys, people, shoreline, and possible sandbars.

This project is to help assist boaters in their navigations through a complex dynamic state of lake and oceanic waters. The expected output will be either bounding boxes or classification labels of the visible hazards. This project will likely use some of the already existing projects such as SeaDronesSee, MODS, MaSTr1325, or LaRS. Specifically MODS has done USV(unmanned surface vehicle) detection. They used COCO/LVIS, which is a common protocol for evaluation of the model. This protocol specifically uses mAP which is the Mean Average Precision. The Mean Average Precision is found by taking the average of the precision recall curve, which is found by adjusting the confidence threshold.

The first possible way to solve this problem is to use YOLO-style object detectors which are bounded boxes and a classification label. This approach is called Yolo because it only looks at the whole image once before predicting the output. “You Only Live Once” (YOLO) encourages taking chances without thinking about the consequences hence the model being named “You Only Look Once”.

The second possible way to solve this problem is to use Faster R-CNN which also uses a bounded box. The main difference is Faster R-CNN uses a two-stage detector. This means the model will first find possible object locations and then try and classify/refine the regions. Contrary to the name, the Faster R-CNN model is slower than the YOLO approach, but it is more accurate.

While MODS looked at YOLO and another two factor model called Mask R-CNN, my approach is using the results found in the MODS to apply specifically to hazard detection systems. Some of the initial image analysis processing steps would include a glare/reflection reduction and edge detection.